

Calculus (PR12)

Week 6: Multivariable Functions and Partial Derivatives

Learning Goals



Multivariable Functions

- Interpret **functions of two variables** and determine **their domains**.
- Visualize and analyze **level curves** and **surfaces**.



Differentiate Multivariable Functions

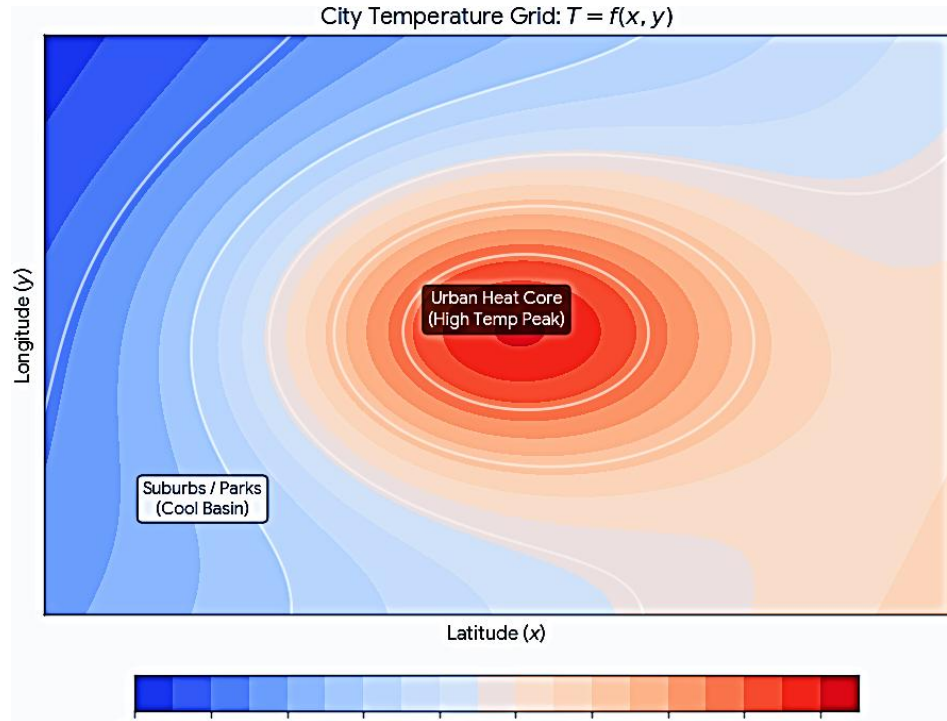
- Compute **partial derivatives** with respect to different variables.
- Compute **higher order** and **mixed partial derivatives**.



Gradients

- Compute and interpret the **gradient vector**.
- Explain the **role of gradients** in **machine learning** and **optimization**.

Seeing Beyond One Dimension



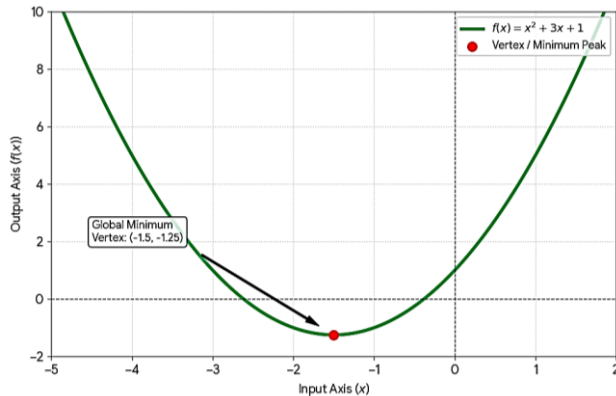
The World is Never One - Dimensional.

Functions of several Variables

From lines to Landscapes:

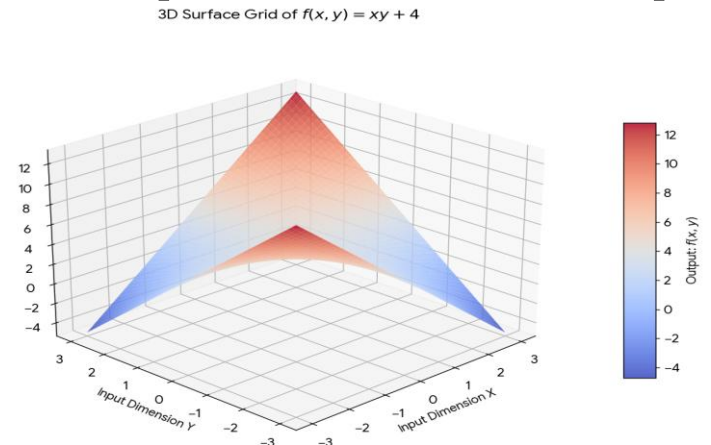
$$y = f(x);$$

One Input $\rightarrow$ One Output



$$Z = f(x, y);$$

Two Inputs $\rightarrow$ One Output



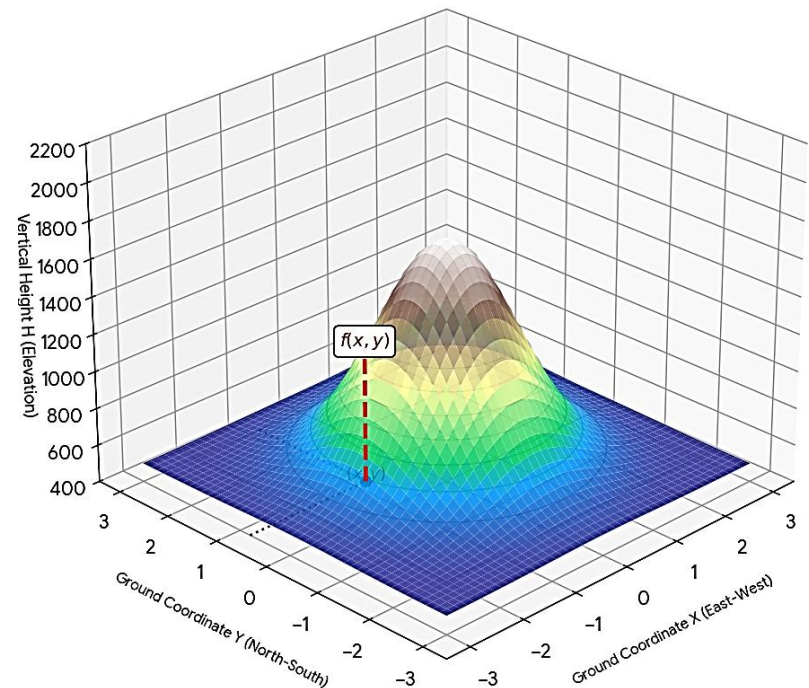
A Mountain Landscape

Imagine standing above a mountain. Every location is defined by:

- East - West position (x)
- North - South position (y)

The height of the mountain is:

$$z = f(x, y);$$



Understanding Domains

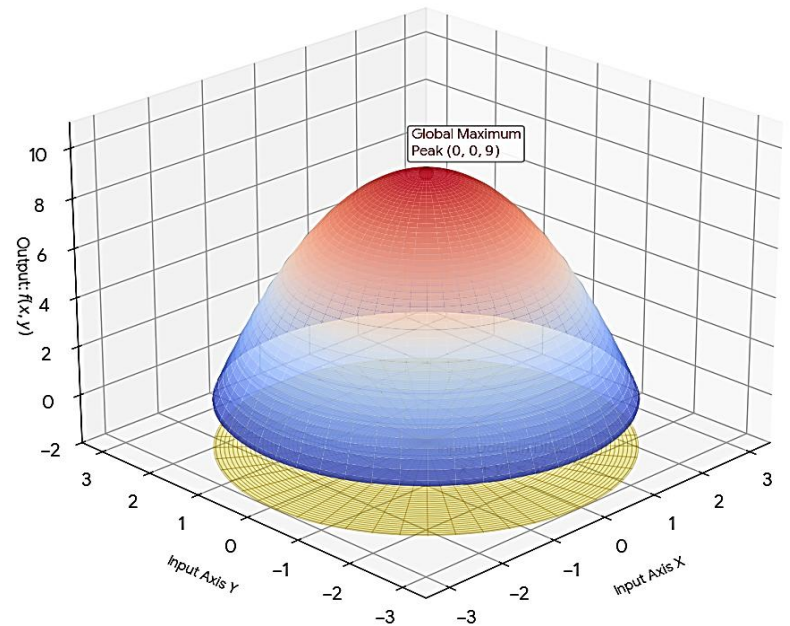
Note: Not every input makes sense.

Example:

$$f(x, y) = \sqrt{9 - x^2 - y^2}$$

For the square root to exist:

$$9 - x^2 - y^2 \geq 0, \text{ and thus } x^2 + y^2 \leq 9$$



Level Curves

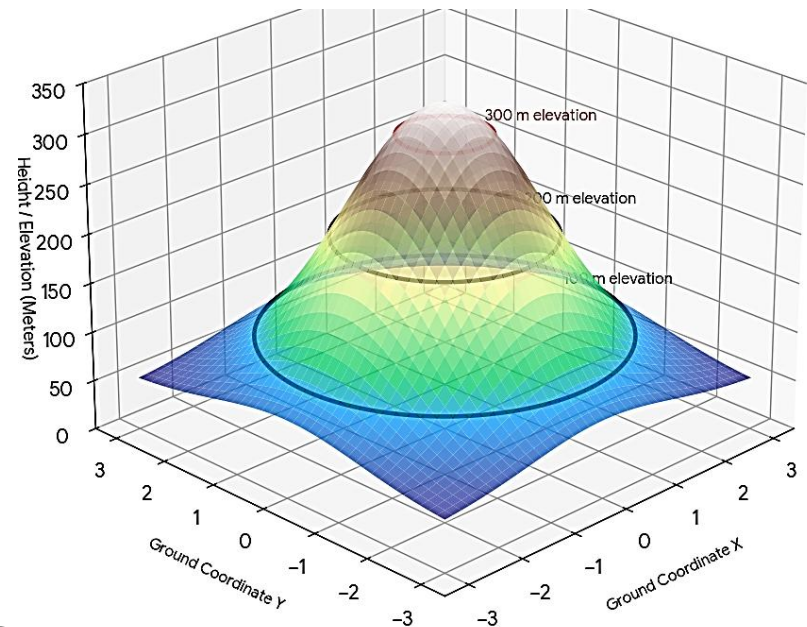
Question: How do we Visualize 3D Functions?
With the Secret Language of Maps.

A hiking map shows:

- **100m elevation**
- **200 m elevation**
- **300 m elevation**

Mathematically:

$$f(x, y) = c$$



AI Interpretation

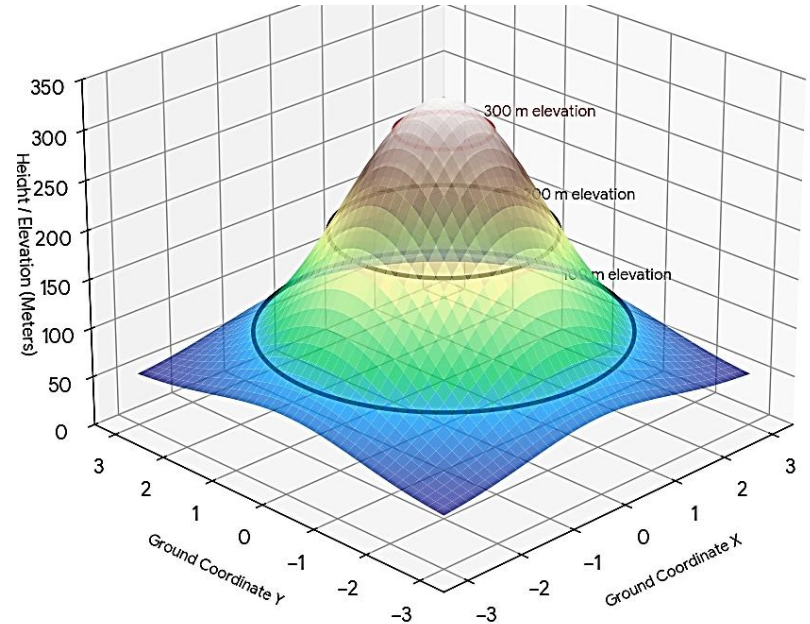
Suppose:

$$f(x, y) = x^2 + y^2$$

Level curves satisfy:

$$x^2 + y^2 = c$$

These are circles.



Question: Why Data Scientists Care?



Partial Derivatives

Intuition: Partial derivatives measure change one variable at a time.

Question: Suppose:

$$f(x, y) = x^2 + 3y$$

How does the output change?

It depends!!!

Partial Derivatives

Definition: A Partial Derivatives measures change with respect to one variable while holding all others fixed.

For:

$$f(x, y) = x^2 + 3y$$

Partial derivative with respect to x :

$$\frac{\partial f}{\partial x} = 2x$$

We treat y as constant.



Partial with Respect to y

Now hold x fixed:

$$f(x, y) = x^2 + 3y$$

Partial derivative with respect to y :

$$\frac{\partial f}{\partial y} = 3$$

We treat x as constant.

DIFFERENT variable, **DIFFERENT** rate of change.

Higher - Order and Mixed PDs

Sometimes we care about how rates themselves change.

$$f(x, y) = x^3 + y^2$$

First partial:

$$f_x = \frac{\partial f}{\partial x} = 3x^2$$

Second partial:

$$f_{xx} = \frac{\partial^2 f}{\partial x} = 6x$$



Mixed PDs - Example

$$f(x, y) = x^2y$$

With respect to x first:

$$f_x = 2xy$$

$$f_{xx} = 2y$$

$$f_{xy} = 2x$$

With respect to y first:

$$f_y = x^2$$

$$f_{yy} = 0$$

$$f_{yx} = 2x$$

The Gradient - The compass of AI

Imagine you are hiking and standing on a mountain.

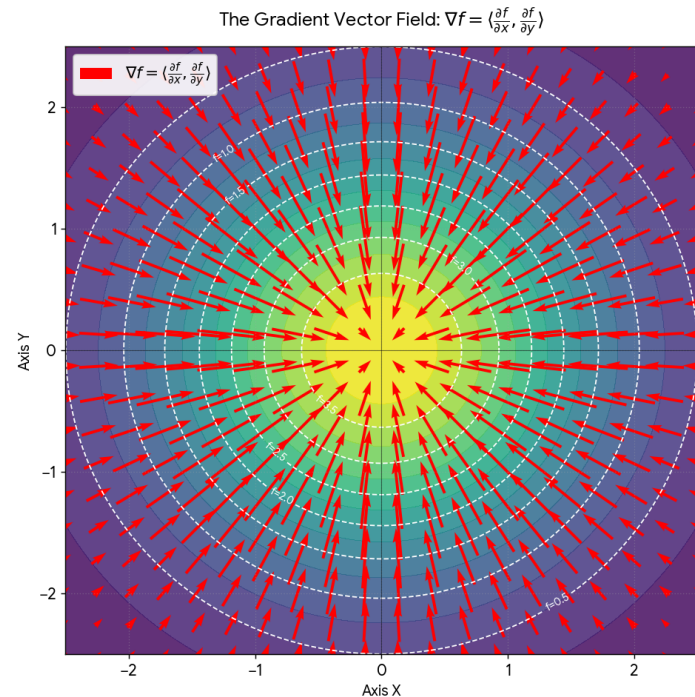


Which direction should you walk?

Definition of Gradient

The gradient combines all partial derivatives:

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$



Note: The gradient is a **VECTOR**, not just a **NUMBER**.

Definition of Gradient

For:

$$f(x, y) = x^2 + y^2$$

$$\nabla f = \langle 2x, 2y \rangle$$

At point (1,2):

$$\nabla f = \langle 2, 4 \rangle$$

So: This vector points toward the direction of greatest increase.



The big Picture

Today we moved from simple curves to multidimensional landscapes. We learned:

- **Functions of Several Variables:** Describe systems with multiple inputs.
- **Domains:** Define where functions make sense.
- **Level Curves:** Reveal the shape of a landscape.
- **Partial Derivatives:** Measure change one variable at a time.
- **Higher - order Derivatives:** Describe curvature and interactions.
- **Gradients:** Provide the direction of steepest increase.

Next week:
Multivariable Optimization.



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Questions

- **Follow-up reading: Calculus: A Complete Course, (sections 13.1 & 13.3 – 13.4 & 13.7)**
- **Joining the Discussion Groups and the Lab**

